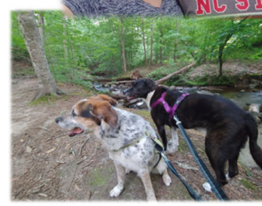
Heat Exchanger Network Design and Operation

Aaron Frye
CHE 450 Lecture
November 14th, 2024

About Me

- Born and raised in North Carolina!
- Career background
  - NCSU CBE Undergrad 2007 – 2012
  - ExxonMobil 2012 – 2021
    - Beaumont Refinery – Utilities, Crude Distillation, Energy
    - Houston Campus – Design, Distillation
    - Baytown Refinery – Crude Distillation, Technical Supervisor
  - NCSU CBE PhD Candidate, Li Research Group 2021 – ???
- Hobbies
  - Disc golf, video games, hiking and backpacking
  - I am married to my lovely wife of 10 years, Rachel
  - We have a dog, Kee'an (white)
  - Rest in Peace to Blaze (black) April 2013 – September 2024



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Learning Objectives

By the end of this discussion, you should be able to:

1) Identify types of heat exchangers and their advantages and disadvantages
2) Conduct heat exchanger network design by using pinch point analysis techniques. Determine the heat exchanger design and hot/cold utility requirements.
3) Apply heat exchanger design principles to analyzing and troubleshooting existing heat exchangers

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Types of Heat Exchangers

Finned Tube

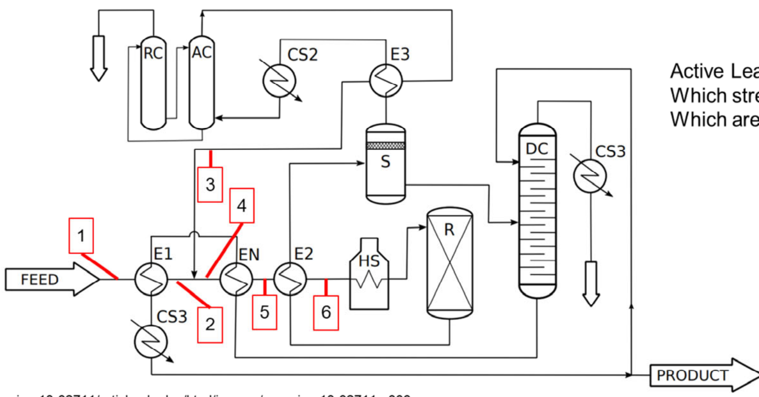
Type	Pros	Cons
Shell and Tube	High area, robust	Cost
Finned Tube	High area	Fouling/Corrosion
Double Pipe	Cheap, simple	Low area
Plate and Frame	Cheap, high area	Mechanical

Double Pipe

Plate and Frame

What is a Heat Exchanger Network (HEN)?

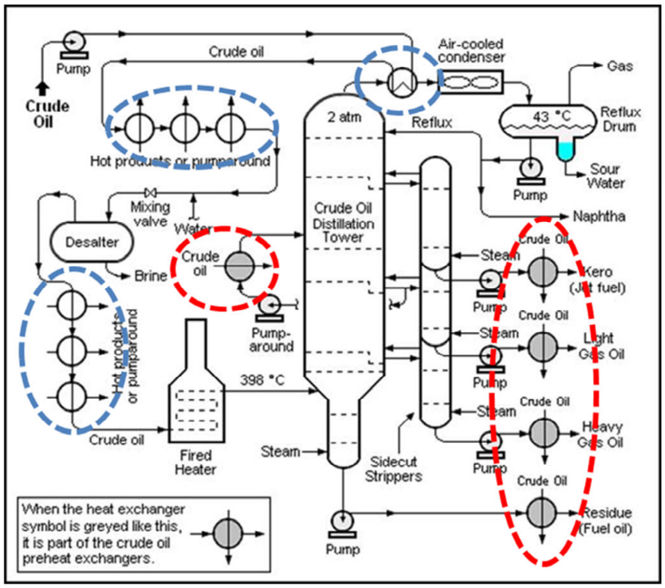
- A series of HX used to recover "waste heat" to heat unit feed (usually)
- "Waste Heat" – heat contained in hot streams which need to be cooled down



Active Learning:
Which stream numbers are hot?
Which are cold?

https://www.mdpi.com/energies/energies-13-02711/article_deploy/html/images/energies-13-02711-g008.png

HENs have many levels of complexity…



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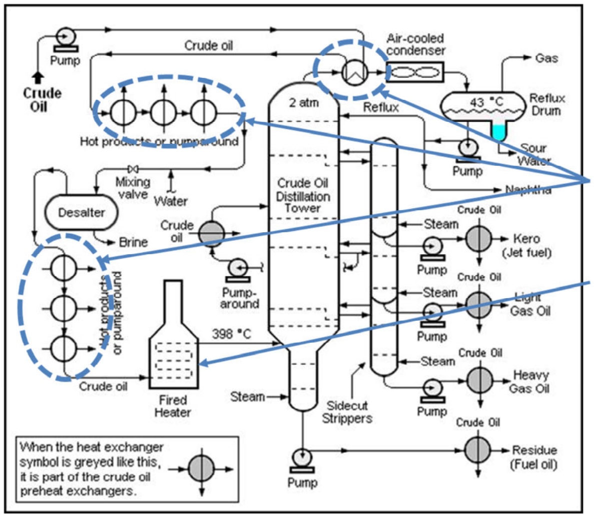
Why do Heat Exchanger Networks matter?



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$$\dot{Q} = \dot{m}C_p\Delta T$$

$\dot{Q}$ = [(200,000 bbl/day)*(42 gal/bbl)*ρ]*C_p*ΔT

Heat exchanger network heats crude from ~50°F to ~500°F = **12,610 MMBTU/day***

Fired heater heats (and boils some) crude oil to ~700°F = **14,400 MMBTU/day**

Active Learning: If energy costs $3/MMBTU, how much is the heat exchanger network worth to your operation each year?

ρ = 0.8 * 8.34lbs H_2O/gal; C_p = 0.5 BTU/lb°F

Approach Concept

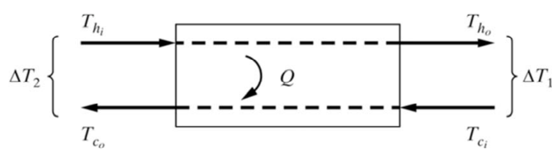
Recall that $\dot{Q} = \dot{m}C_p\Delta T$ and $\dot{Q} = U * A * LMTD$ where $LMTD = \dfrac{\Delta T_2 - \Delta T_1}{(\ln(\Delta T_2/\,\Delta T_1))}$

For both, ΔT is the driving force!! "Approach" is the term for this driving force

Temperature crossover is physically impossible (i.e. if ΔT goes to zero, Q is zero)

What would you say is a practical approach ΔT?

ΔT_2 Cold out Approach



ΔT_1 Cold in Approach

Typical Design Algorithm:

1. Collect process data (or desired design specs) such as flow rates, temperatures, heat capacity
2. Perform Pinch Analysis (temperature interval diagram, cascade analysis)
3. Initially match hot and cold streams using the pinch point and some logic
4. Design the exchangers

Heat Exchanger Network Design

1. Collect Process Data

Stream	m (kg/s)	T_{in} (ºC)	T_{out} (ºC)	mC_p (kJ/s.ºC)
H_1	10	180	60	3
H_2	25	150	30	1
C_1	12	30	135	2
C_2	17	80	140	5

If we were to cool and heat the streams independently for 1 year (the 2023 cost for industrial electricity is ¢7.91/kWh), we would spend **$519,685** for cooling and **$584,818** for heating.

Acknowledgement: Thank you to Dr. E. Daniel Cárdenas-Vásquez for developing the pinch analysis problem

Pinch Analysis – Handouts Time!

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Picking back up from the handouts

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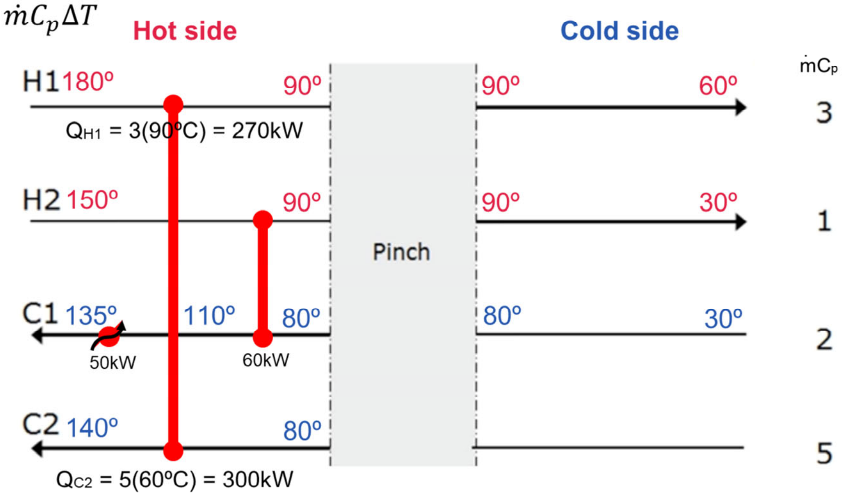
Heat Exchanger Network Design

4. Design the Exchangers

$$\dot{Q} = \dot{m}C_p\Delta T$$



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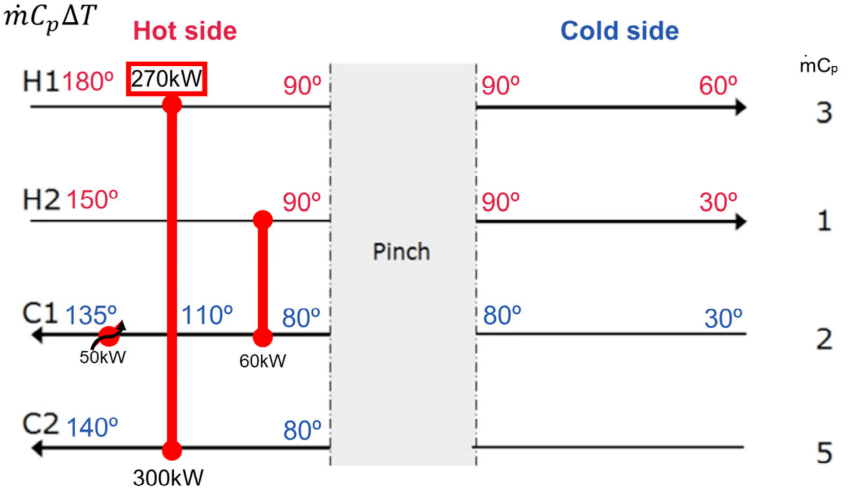

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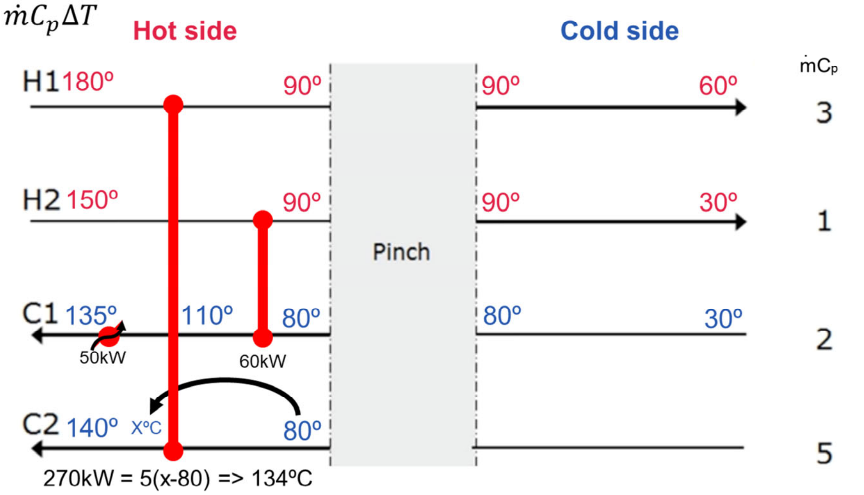

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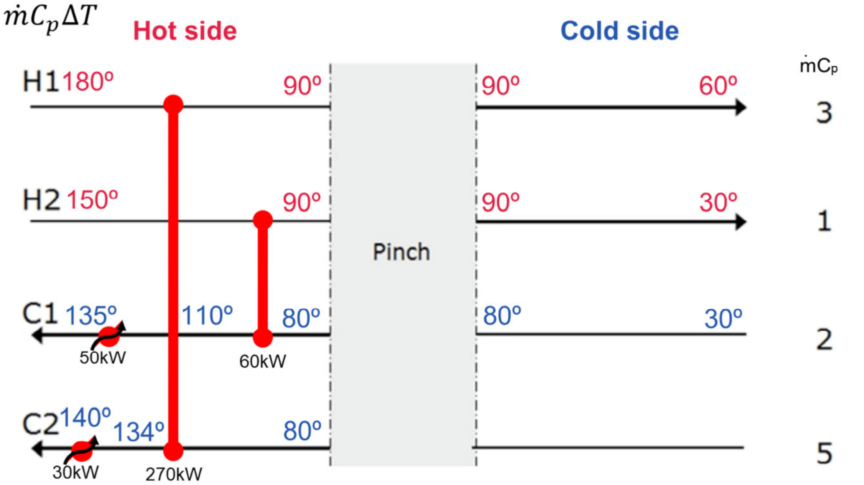

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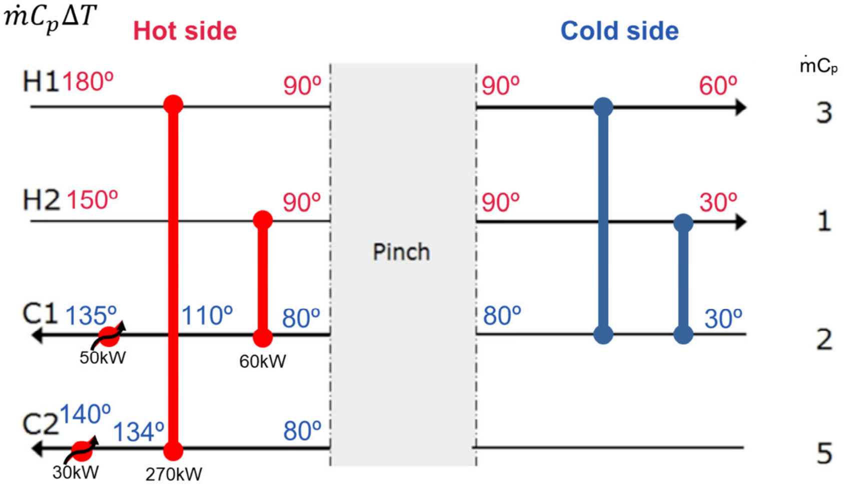

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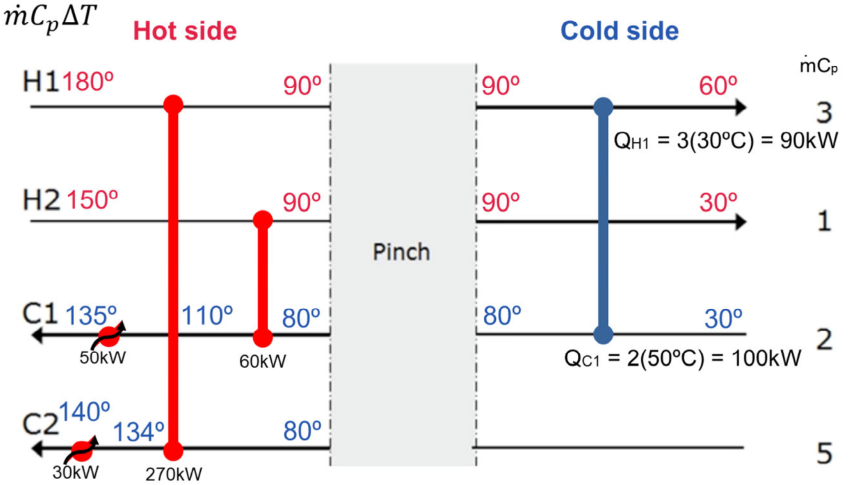

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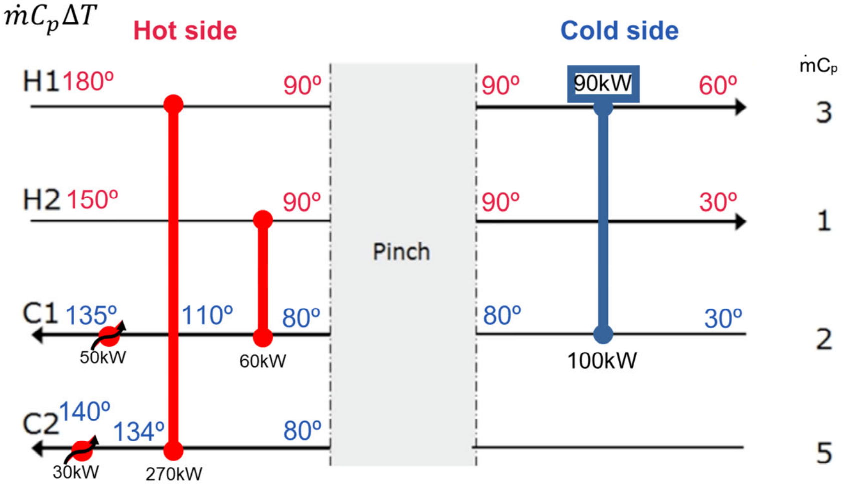

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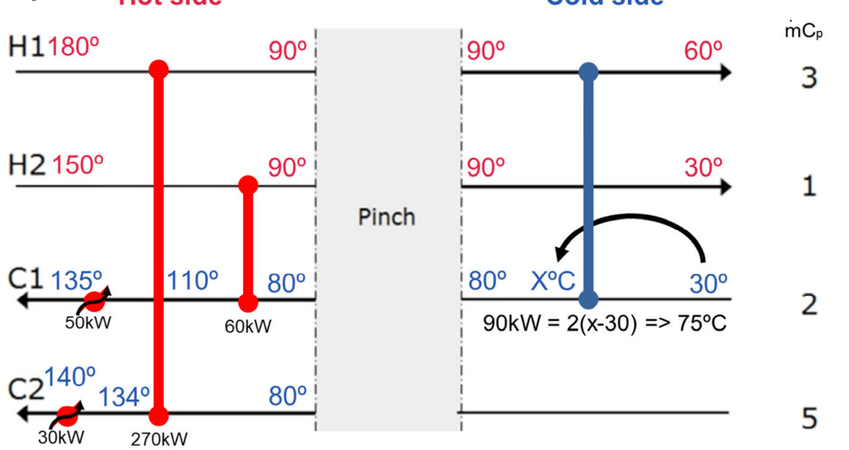

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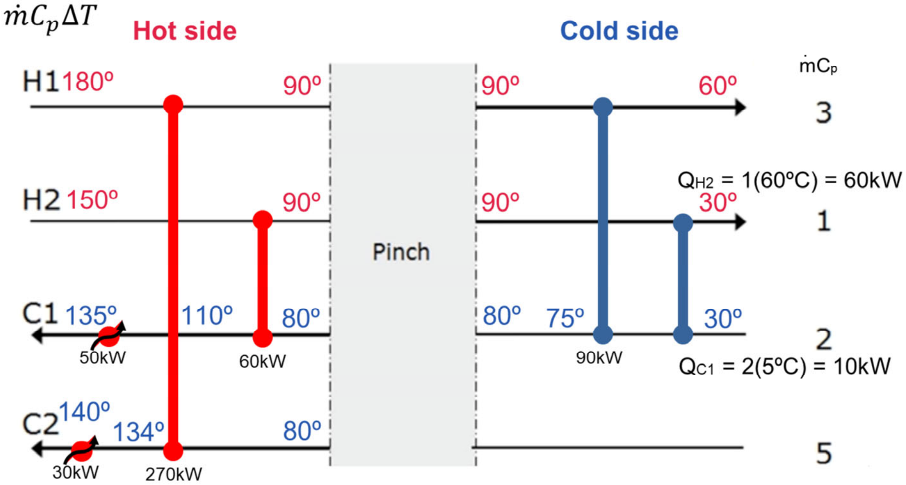

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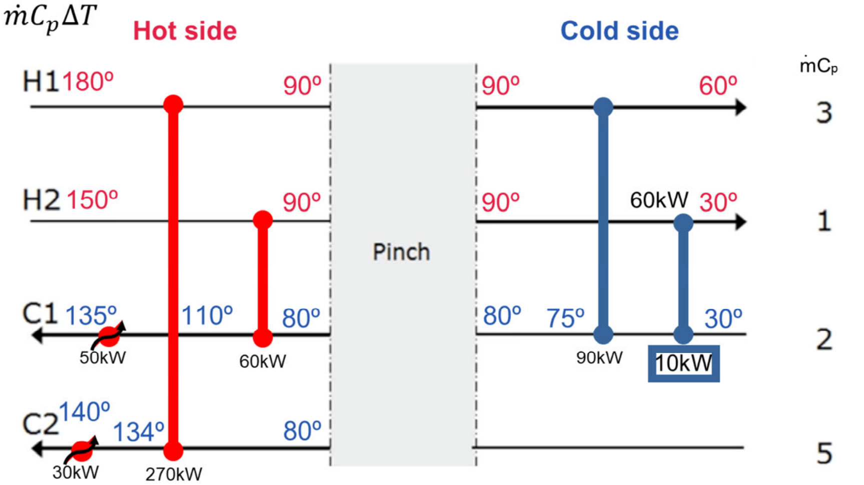

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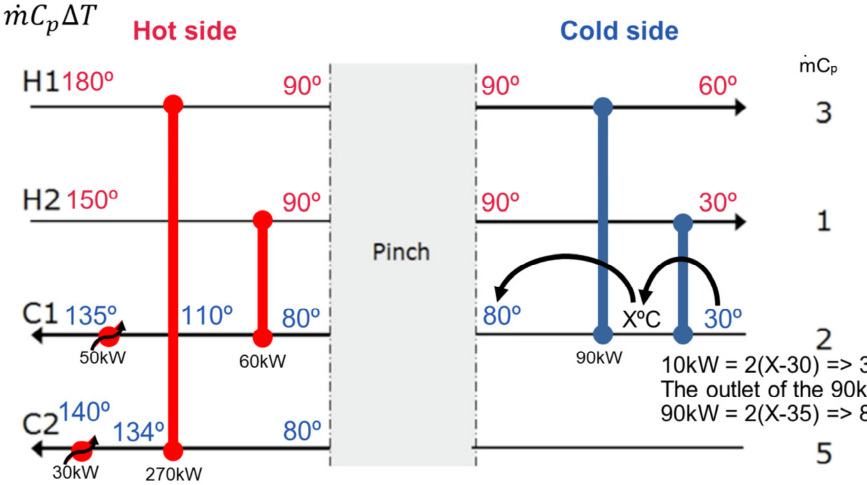

10kW = 2(X-30) => 35ºC
The outlet of the 90kW HX is now:
90kW = 2(X-35) => 80ºC!

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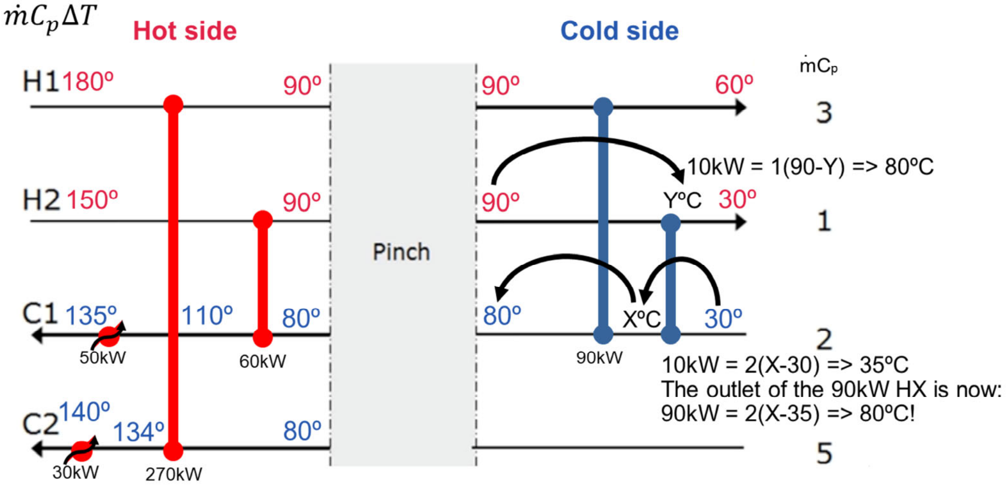

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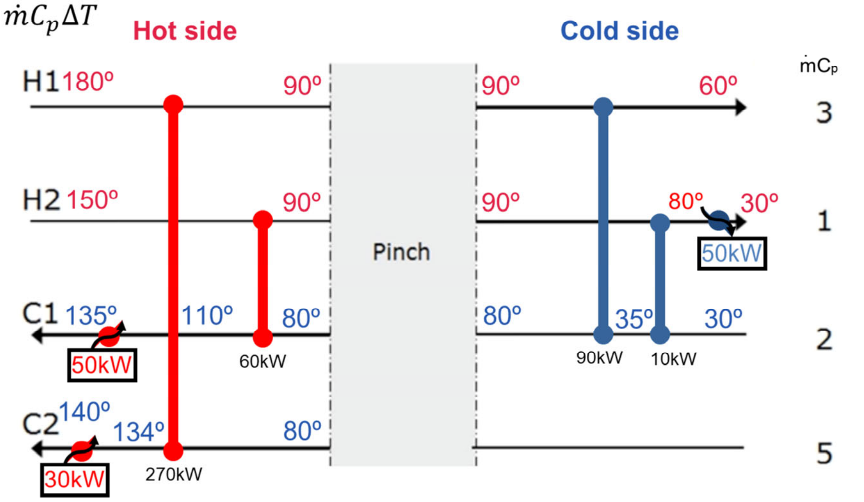

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Good designs save money!

If we were to **cool and heat the streams using a heat exchanger network** for 1 year (the 2023 cost for industrial electricity is ¢7.91/kWh), we would spend **$61,307** **for cooling** (11% of original) and **$91,738** **for heating** (15% of original)

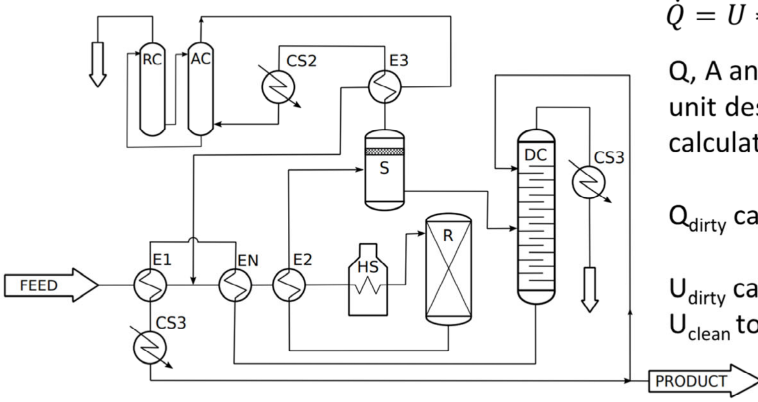
CHE450 covers DESIGN
…but what if the HEN already exists?

Tools at your disposal for existing equipment

- Process data – temperatures, flows
- Design basis – what should the equipment achieve were it functioning properly? This most closely ties back to CHE450
- Operator input – cause/effect, how it normally works

Assume this unit exists

How do you know which exchanger is fouled?



$\dot{Q} = U * A * LMTD$

Q, A and LMTD are available from the unit design basis which allows you to calculate U_{clean} (and Q_{clean})

Q_{dirty} can be calculated from $\dot{Q} = \dot{m}C_p\Delta T$

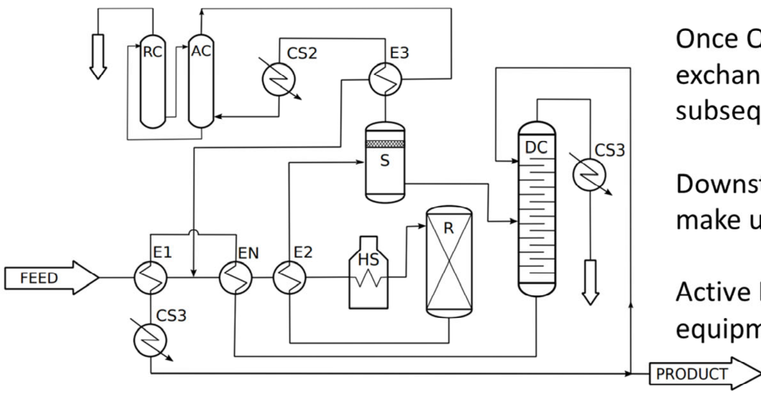
U_{dirty} can then be found and compared to U_{clean} to determine "degree of fouling"

https://www.mdpi.com/energies/energies-13-02711/article_deploy/html/images/energies-13-02711-g008.png

Assume this unit exists

What will happen if you take it out of service for cleaning?

Even a fouled exchanger is still contributing Q_{dirty}



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It gets complicated…

Once Q_{dirty} is lost, downstream exchangers will gain approach (and subsequently duty)

Downstream exchangers alone will not make up for Q_{dirty}

Active Learning: which piece of equipment makes up for the lost duty?

Any Questions?

Thank you!